

COURSE SYLLABUS

Introduction to Computer Science

Course Code: CS101

Instructor: Dr. Jane Smith

Credit Hours: 3

Course Description

This course introduces fundamental concepts in computer science including algorithms, data structures, and programming. Students will learn problem-solving techniques and write programs in Python.

Course Objectives

1. Understand fundamental programming concepts
2. Learn data structures and algorithms
3. Develop problem-solving skills
4. Practice software development best practices

Topics Covered

Module 1: Introduction to Programming - Variables, data types, loops, conditionals

Module 2: Functions and Modular Design - Function definition, scope, recursion

Module 3: Data Structures - Lists, tuples, dictionaries, sets

Module 4: Algorithms - Sorting, searching, complexity analysis

Module 5: Object-Oriented Programming - Classes, inheritance, polymorphism

Assessment

Quizzes: 20%, Assignments: 30%, Midterm: 20%, Final Project: 30%